

Unit III Convolutional Neural Networks

Unit III – Convolutional Neural Networks (CNN)

1. CNN Kya Hai?

CNN ek special deep learning architecture hai jo image aur spatial data ke liye use hoti hai. Isme features ko automatically learn kiya jata hai.

Why CNN Is Powerful

- ? Spatial relationships ko samajhne mein helpful
- ? Image data ke liye efficient
- ? Feature extraction automatically hota hai

2. Convolution Layer

Convolution layer input image par filters apply karta hai. Ye edges, textures aur patterns detect karta hai.

Key Idea

Filter input par slide hota hai aur dot product compute karta hai.

3. Padding

Padding edges par zeros add karta hai taki output size preserve ho sake.

Types

- ? Valid padding
- ? Same padding

4. Stride

Stride batata hai ki filter kitni step mein move karta hai.

Effect

? Stride bada => output size kam

? Stride chhota => output size bada

5. Pooling Layer

Pooling layer spatial dimensions reduce karta hai. Ye computation ko kam karta hai aur overfitting ko reduce karta hai.

Common Types

? Max Pooling

? Average Pooling

6. Flattening

Flattening 2D feature maps ko 1D vector mein convert karta hai. Iske baad dense layer mein input diya jata hai.

7. Dense Layer

Dense layer fully connected layer hoti hai. Ye final classification ya regression ke liye use hoti hai.

8. 1x1 Convolution

1x1 convolution ka use channels ko change karne aur network ko efficient banane ke liye hota hai.

9. Input Channels

RGB images mein usually 3 channels hote hain: Red, Green, Blue. CNN in channels par operate karti hai.

10. Inception Network

Inception network multiple filter sizes ko ek saath use karta hai. Ye feature extraction ko better banata hai.

11. Transfer Learning

Ek pre-trained model ko dusre task ke liye reuse karna transfer learning kehlata hai.

Benefits

- ? Time save hota hai
- ? Data requirements kam hote hain
- ? Better performance mil sakta hai

12. One Shot Learning

Isme bahut kam examples se learning ki jati hai. Ye face recognition jaise tasks mein useful hota hai.

13. Dimension Reduction

CNN mein features ko reduce karke model ko simple aur efficient banana possible hota hai. PCA aur autoencoders isme help kar sakte hain.

14. CNN Applications

CNN ka use aaj kal in areas mein hota hai:

- ? Image classification
- ? Object detection
- ? Face recognition
- ? Medical imaging
- ? Autonomous vehicles

15. Important Exam Points

- ? CNN images ke liye powerful deep learning model hai.
- ? Convolution, pooling aur flattening basic CNN concepts hain.
- ? Padding aur stride output size ko control karte hain.
- ? Transfer learning bahut practical aur important topic hai.